

ACT-FP-5500 'Liberty' Composite Frac Plug, 5-1/2 in.

Technical Datasheet DS-FP-5500 Rev E | Product family: ACT-FP-5500

1. Product Overview

The ACT-FP-5500 'Liberty' composite frac plug enables plug-and-perf multistage stimulation in 5-1/2 in. casing. Filament-wound glass/epoxy construction with one-piece cast iron slips mills out in under 12 minutes average. A dissolvable magnesium frac ball is standard; a phenolic ball option is available for low-temperature wells. Qualified to API 19V V3.

2. Applications

- Plug-and-perf horizontal completions
- Zonal isolation during hydraulic fracturing
- Re-frac isolation

3. Features

- 10,000 psi differential at 300 degF
- Average mill-out time 11.4 min (field data, n=612 plugs)
- Dissolvable Mg alloy ball standard (1.75 in.)
- 8 brass shear screws - 55,000 lbf setting force (increased from 6 screws per ECO-2025-014)
- Runs on ACT-ST-10 wireline setting tool with FP-AK-55 kit
- Ceramic-button one-piece slips prevent pre-set and casing damage
- API 19V Grade V3 qualified

4. Technical Specifications

Parameter	Specification
Casing Size	5-1/2 in., 17 - 23 lb/ft
Maximum OD	4.312 in.
Overall Length	18.9 in.
Differential Pressure Rating	10,000 psi
Temperature Rating	300 degF (HNBR)
Frac Ball	1.75 in. dissolvable Mg alloy (std); phenolic option -08-P
Ball Seat ID	1.42 in.
Setting Force	55,000 lbf (8 x 1/4-28 brass shear screws)
Setting Tool	ACT-ST-10 (#10) with ACT-FP-AK-55 adapter kit
Max Setting Depth / Deviation	Tested to 89 deg deviation
Qualification	API 19V Grade V3

All ratings are at ambient temperature unless otherwise noted. Performance envelope per API 11D1 V3 qualification testing where indicated. Consult ACT Engineering for combined load conditions.

5. Materials of Construction

Component	Material
Mandrel, cones, guide, seat	Filament-wound glass/epoxy composite
Slips	Cast iron Class 30 with ceramic buttons (9 per slip)
Element	HNBR 90A, compound ACT-H90
Shear screws	Brass C36000, 1/4-28 UNF
Frac ball	Dissolvable Mg alloy (std)

6. Ordering & Reference Documents

Order by part number and specify: casing size and weight, service trim (standard or NACE sour), elastomer option, end connections, and any customer inspection requirements. Related documents:

- BOM-FP-5500 Rev E
- MAN-FP-5500 Manual
- ECO-2025-014
- QIR-2024-0871 (composite incoming inspection)

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